

AIC and ANOVA

June 2025

What to do with our model set?

- We were excellent planners
 - Developed clear hypotheses
 - Model set checked and ready to go
- Now what?
- Choose the "best" model
 - AIC
 - ANOVA

Crime

Does education help explain the rate of violent crime?

- `Total`: the total number of crimes per million
- `Violent`: the number of violent crimes per 100000
- `Police`: Police funding in dollars per resident per year
- `Highschool`: Percent of adults with 4 years of high school
- `Delinquent`: Percent of 16-19 years olds not in high school
- `College`: Percent of 18-24 year olds in college
- `CollegeAdults`: Percent of adults with at least 4 years of college

What's our plan?

$$n/15 = 50/15 = 3.33$$

- Not going to use all the predictors
- Construct hypotheses
- Control all hypotheses for the amount of money spent on policing
 - Hypothesis #1:
 - Low education levels are a the driver of violent crime

```
crime1 <- glm(Violent~Police+HighSchool+Deliquent,family = poisson(), data=crime)
```

- Hypothesis #2:
 - More college educated adults means less violent crime

```
crime2 <- glm(Violent~Police+CollegeAdults,family=poisson(),data=crime)
```

Coding Time

This is a continuation of our planning activity

- Make additions and modifications to the previous Quarto file
- Keep the same initial model
- Develop, fit and assess a second model hypothesis

Choices, choices

- Akaike's Information Criterion (AIC)
 - Penalizes for the number of estimated parameters

$$-2\log - \textit{likelihood} + 2p$$

- Roughly the same sample size between models being compared
- Measures the information lost
 - Goal is the least amount of lost information
- Lowest AIC = "best" model
 - Marked preference when the difference between AIC values is greater than 2
 - Subjective
- Picks the "best" model in the set
 - Doesn't mean its any good
 - A better one could always exist

Different Flavours of AIC

- Correct for small sample size, AICc

$$-2\log - \text{likelihood} + 2p + 2\frac{p(p+1)}{(n-p-1)}$$

- Handle over-dispersed data, QAIC

$$QAIC = -2\frac{2}{VIF}\log - \text{likelihood} + 2p$$

- Can do QAICc too!
- Measures relative, not absolute, fit

Finding AIC

AIC in R

```
AIC(crime1, crime2)
```

```
##           df      AIC  
## crime1    4 13093.09  
## crime2    3 14780.93
```

The model with the covariates related to high school education performs better

WARNING!

AIC is only defined up to an arbitrary constant

- Different constants for different purposes
- Take care when comparing fits of different classes

Coding Time

Use AIC to select between the two models you defined earlier

Analysis of Variance (ANOVA)

Is one of our models doing a better job explaining the variability in the data?

Variability is divided into two parts:

- Explained by the model - Regression sums of squares
- Unexplained by the model - Residual sums of squares

How's it work?

Models with more explanatory variables should explain more variability

- If true, keep the model with more covariates
- If the models explain a similar amount of variability, keep the model with fewer covariates

H_0 : The two models are equivalent

H_1 : The model with more covariates explains more variability

ANOVA

```
anova(crime1, crime2)
```

```
## Analysis of Deviance Table
##
## Model 1: Violent ~ Police + HighSchool + Deliquent
## Model 2: Violent ~ Police + CollegeAdults
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1         46      12691
## 2         47      14381 -1  -1689.8 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Strong evidence against the null hypothesis

Null Model

Do we learn anything at all from doing regression?

- ANOVA to compare full and null model
 - Full model: the one with explanatory variables
 - Null model: intercept only
 - The hypothesis test at the end of the `lm` output

```
crime.null <- glm(Violent~1, family=poisson(), data=crime)
anova(crime1,crime.null)
```

```
## Analysis of Deviance Table
##
## Model 1: Violent ~ Police + HighSchool + Deliquent
## Model 2: Violent ~ 1
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1         46      12691
## 2         49      20264 -3   -7573.1 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Now you try

- Compare your two models using ANOVA
- Build a null model and compare to the chosen model using ANOVA